PV · BATTERY · WALLBOX · HEAT PUMP

AI is smart enough.
What it lacks is context.

OnlyPanels gathers every fact about a home it can find, with or without the owner's address, so the first design is one an expert would sign.

WHAT WE FOUND IN THE DATA

Same request, different offer.

59%

of solar projects share their exact request with another customer: every field the same

99%

of those identical requests still got different designs from the experts

6.3 kWp

average solar spread inside one identical request: a whole roof side

The experts are not random. They knew things the request did not say: the roof, the shade, the street. Half of the request fields are empty.

THE INSIGHT

The model is not the bottleneck. The data is.

- **Homeowners leave fields empty.** They don't know their roof slope, or don't want to share where they live. That is their right.
- **So a bigger model learns noise.** On past projects any model is only a little better than one design for every home.
- **We respect the choice and still do better:** extract everything we can from what is shared, and make every assumption visible.

ONE CLICK ON A HOUSE

Twelve sources, fused into one picture of the home.

3D roofs
Bavaria LoD2: every roof side, slope, direction, area

Aerial photo
Bavaria DOP20, 20 cm, on ground and roofs

Sun per surface
EU PVGIS, real hourly weather years

Tree shade
Bavaria's single-tree map, lowers yield per side

Census 2022
Building age and heating in the neighbourhood

Heat plan
Munich: is district heating coming here?

Monument map
Bavarian listed buildings: what is allowed

Address
Photon / OpenStreetMap geocoding

Roof photo check
Claude vision finds windows, chimneys, old panels

Price book
The planner's own prices, parts and labour

Rules and money
Local power price, feed-in, KfW and Munich grants, §14a

1,041 past projects
How this installer sizes, learned per installer

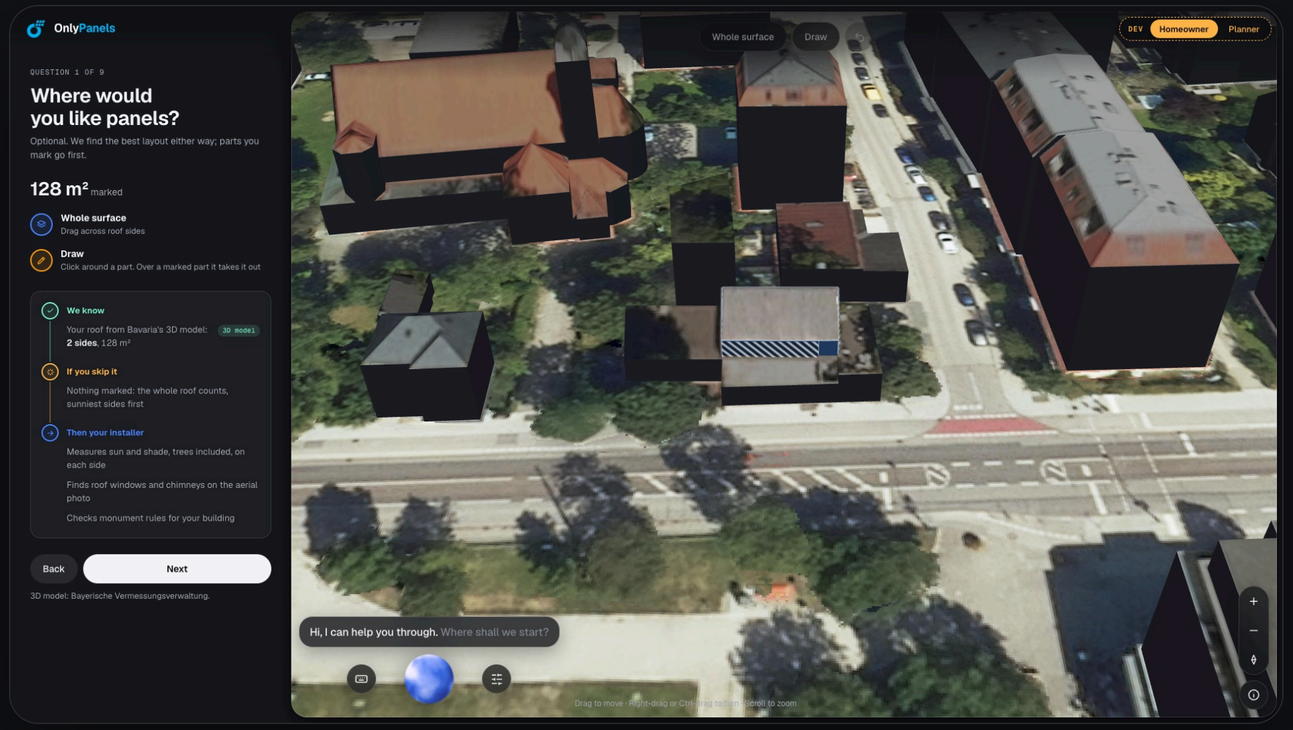
open public data · AI vision · installer and law


OnlyPanels
DEV
Homeowner
Planner
Agnes-Bernauer-Straße 98, 80687 Munich
YOUR REQUESTS
Am Blütenring 42, 80939 Munich
Sent
R-9PX4N · sent 30 min ago
Sondermeierstraße 86, 80939 Munich
Sent
R-3KD8W · sent 2 h ago
Sondermeierstraße 74, 80939 Munich
Sent
R-7MQ2A · sent 5 h ago
Agnes-Bernauer-Straße
Lassstraße
LATIM
THE HOMEOWNER'S MAP
Munich in 3D, every roof measured,
on the real aerial photo.
Hi, I can help
Find your area, then click the buildings you want
With map
Without map

RESPECT FIRST

Share less, still get a real offer.

- **Every question optional.** Under each: what we know, what we assume if skipped, what the installer does.
- **Nothing leaves the browser before Send.** The home is worked out in the page.
- **The homeowner picks what goes.** No map? Words like "south roof, 40 m²" still size the system.



THE PLANNER'S INBOX

A full draft is waiting before anyone opens the request.



Home loaded → trees checked → roof space → system sized → rules checked → priced. About a second, every step with what it found.

THE PLANNER'S DESK

Every fact has a source. Every number has an owner.

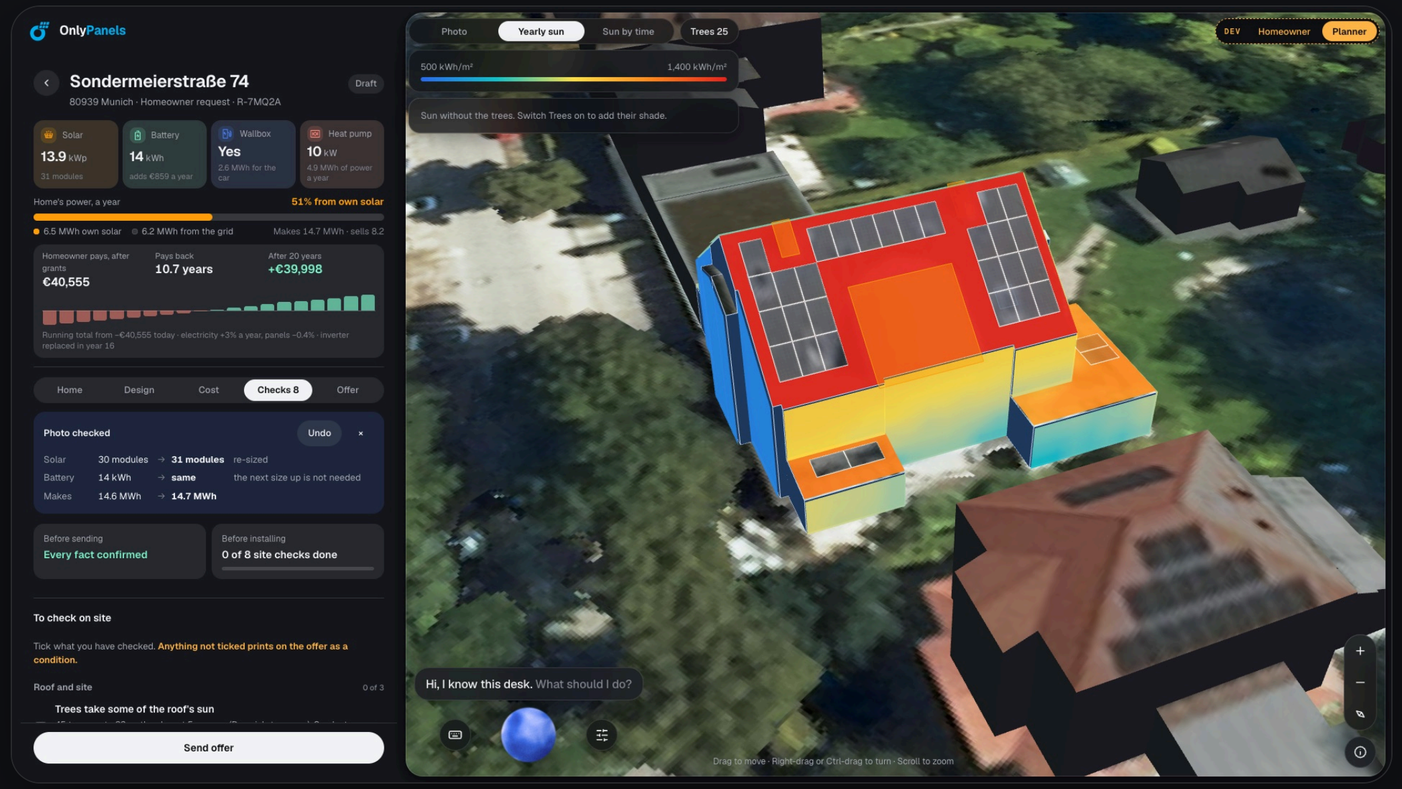
- **Facts tagged** customer, measured, found or assumed. Assumed facts block Send until a person confirms.
- **The engine decides the numbers,** never the language model.
- **The review reruns the engine** at the edges of each guess: "ask about the bill, not the car".



SUN, SHADE, OBSTACLES

Panels go where the sun is, not where the windows are.

- **Yearly sun per roof side**, with tree shade taken off.
- **Roof windows and chimneys** found on the aerial photo; no module touches one.
- **A weaker roof gets more modules,** a stronger one fewer, as a planner would.



MONEY AND LAW

A price the homeowner can act on.

- **Bill of materials** from the installer's own price book.
- **Grants and rules for this home:** KfW, Munich's heat pump grant, negative-price hours, smart meter, §14a.
- **Payback and 20 years,** the same figure for planner and homeowner.



MEASURED HONESTLY

On 276 projects the engine never saw

	ENGINE	SAME DESIGN FOR ALL	RULE OF THUMB
Solar, mean error	**2.65 kWp**	3.06 kWp	4.02 kWp
Solar within 20% of the expert	**49%**	44%	31%
Battery, mean error	**2.66 kWh**	3.03 kWh	3.56 kWh

Past projects have no roof, no address, no shade, so this is the engine at its weakest. The gap that remains is exactly the data OnlyPanels now collects. It behaves like a planner: a car never shrinks solar, more use never shrinks the battery, the roof limit always holds.

Split by customer, results per installer in docs/data/findings.md · scripts/final_test.py

HOW WE BUILT IT

Tech stack

Engine and API

Python, FastAPI, scikit-learn models per installer, hourly energy simulation on PVGIS weather years, rules for heat pump, grants and law

Web

React 19, TypeScript, Vite, MapLibre GL, three.js for 3D roofs and modules, web workers for city-scale geometry

AI

Claude (Anthropic) for roof photo vision, the planner's sentence box and voice agent via MCP tools. **ElevenLabs** for speech in and out: the planner runs the desk by voice

Bavaria LoD2 · DOP20 · Einzelbäume · PVGIS · Zensus 2022 · Wärmeplan München · Denkmal · Photon / OSM

OnlyPanels

Better facts, not a bigger model.

A first design and its price in minutes. The planner stays in control, the homeowner decides what is shared, and every number can be traced to where it came from.

github.com/atahankilc/onlypanels